

Speak Like A Mathematician:

slope:

parallel lines (slopes):

perpendicular lines (slopes):

coinciding lines:

Slopes of Parallel and Perpendicular Lines

Two nonvertical lines are parallel if and only if they have the _____ slope.

Two vertical lines are always _____.

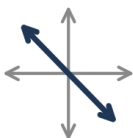
Two nonvertical lines are perpendicular if and only if their slopes are _____ (their product is _____).

A _____ line and a _____ line are always perpendicular to each other.

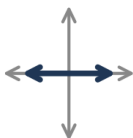
Positive



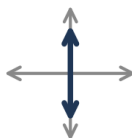
Negative



Zero



Undefined



Forms of a Linear Equation

Point-slope form: _____ (uses a point (x_1, y_1) and slope m)

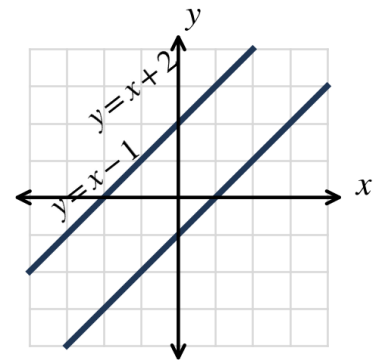
Slope-intercept form: _____ (uses the y-intercept b and slope m)

Vertical line: _____ Horizontal line: _____

Example 1 - Find the Slope of a Line

Find the slope of the line through each pair of points.

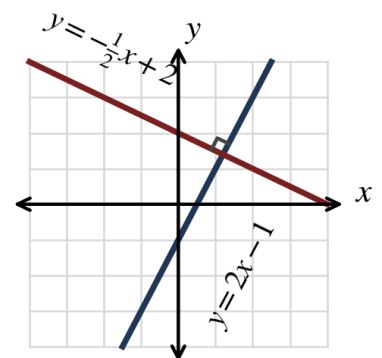
- a) $(6, -1)$ and $(-2, 5)$
- b) $(-5, 2)$ and $(7, 8)$
- c) $(4, -3)$ and $(4, 9)$
- d) $(-6, 5)$ and $(2, 5)$



Example 2 - Find the Missing Coordinate

Use the slope formula to find the missing coordinate.

- a) $m = -2/3$ through $(-4, 5)$ and $(2, \underline{\hspace{1cm}})$
- b) $m = 3/4$ through $(1, -2)$ and $(9, \underline{\hspace{1cm}})$
- c) $m = -1/2$ through $(\underline{\hspace{1cm}}, 7)$ and $(10, 1)$



Example 3 - Write an Equation of a Parallel Line

Write an equation of the line that passes through $(-2, 5)$ and is parallel to $y = 3x - 4$.

- a) Find the slope of the given line.
- b) Write the equation in point-slope form.
- c) Write the equation in slope-intercept form. Then check your answer.

Example 4 - Write an Equation of a Perpendicular Line

Write an equation of the line that passes through $(4, -1)$ and is perpendicular to $3x + y = 7$.

- a) Write $3x + y = 7$ in slope-intercept form.
- b) Find the slope of the perpendicular line.
- c) Write the equation. Then check your answer.

Example 5 - Parallel, Intersecting, or Coinciding?

Determine whether the lines are parallel, intersecting, or coinciding.

- a) $y = 4x - 3$ and $y = 4x + 1$
- b) $y = -2x + 5$ and $4x + 2y = 10$
- c) $y = (1/2)x + 6$ and $y = -3x + 6$
- d) $3x - y = 8$ and $x + 3y = 15$